



MEGR 3156

Formal Design Review

Design Team 7

Camden Bowers, Angie Gonzalez-Lopez, Timothy Mitchell, Derek Penland, Tristan Ray, Miguel Salas, Evelyn Wagner

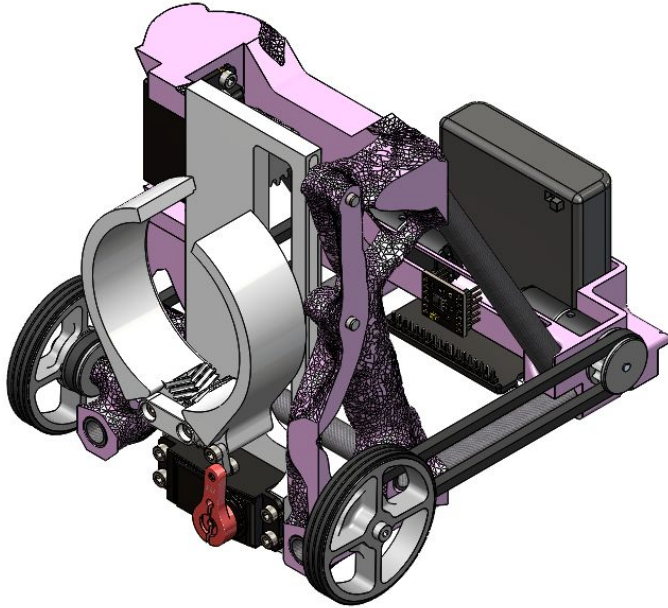




UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Overall Design

Overall Design



- Two piece PETG Filament chassis with connecting carbon fiber rods
- Rack and Pinion lift system
- Gripper actuated by a single servo and gear mechanism with servo lockout utilizing linkage geometry
- Belt drive using O'Rings for traction
- Rear mounted motors and battery for CG
- Optimized to store the can as far as possible inside of the robot



UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Gripper Mechanism

Gripper

Claw Gripper System with Single Servo

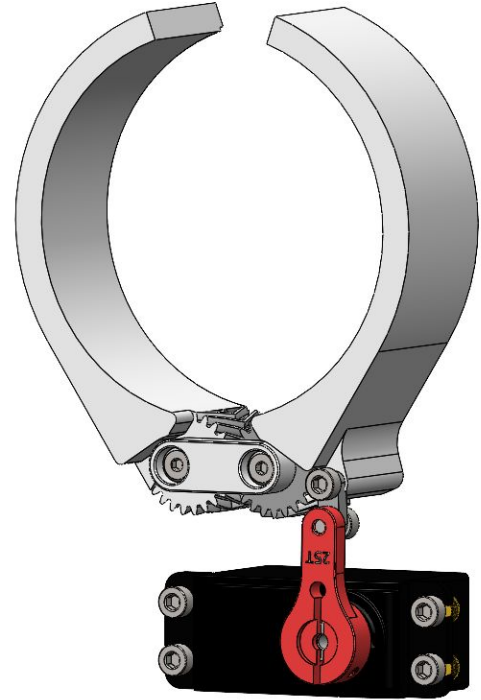
Servo: INJORA INJS022 22KG Low Profile
180° Digital Servo

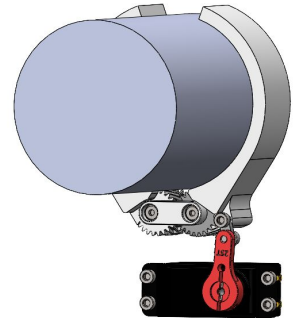
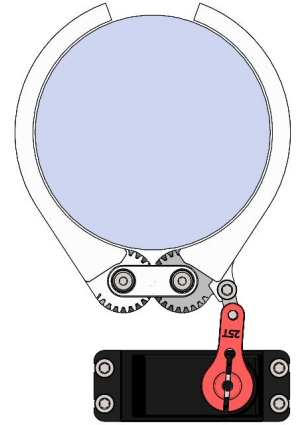
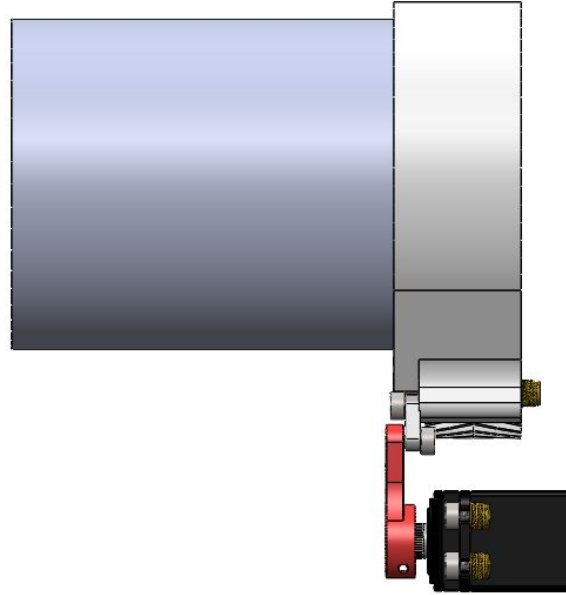
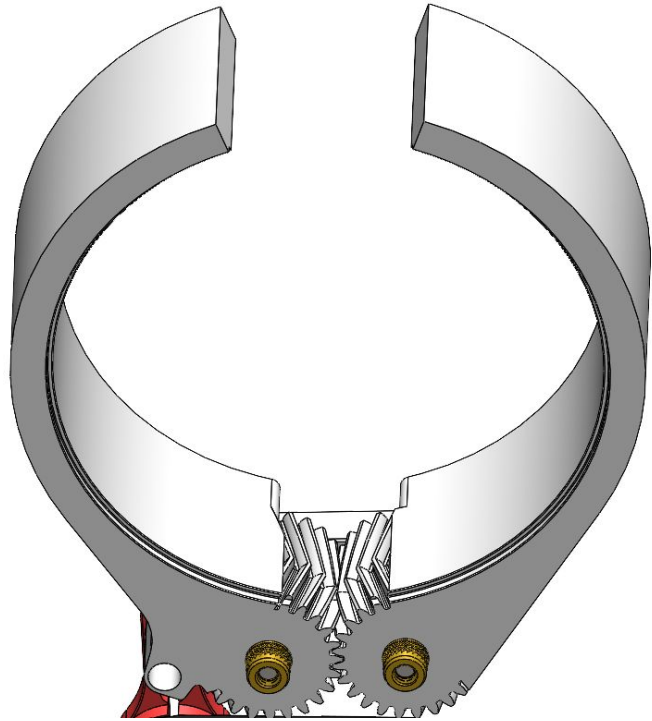
Servo Lockout Linkage

Material: PLA Filament

Total Weight (including hardware): 80g

Claw Deflection: 2.97 μm







UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Lift System

Lift System

Rack and pinion lift system:

Pressure angle (deg) – 14.5°

Face width – 10mm

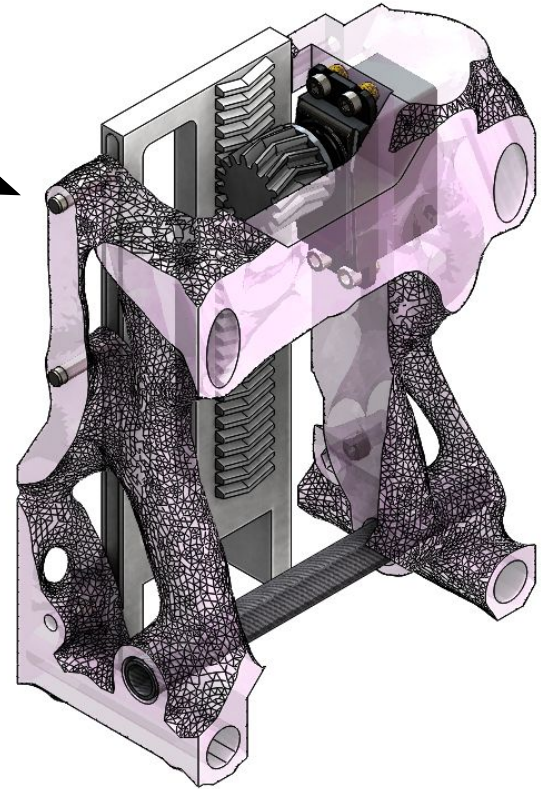
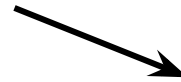
Servo: INJORA INJS022 22KG Low

Material: PLA Filament

Total weight (including hardware): 92.03g

Double Herringbone Gear Configuration

Rack retention with 4
Dowel Pins



Lift System

The rack is 100 mm in length.

Pinion radius 15mm

Designed for 16.5 N upward load

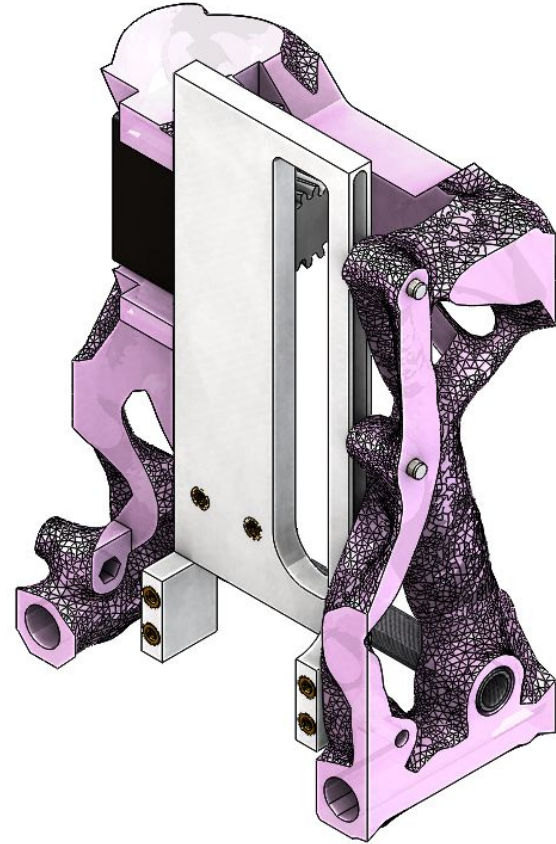
Safety factor > 3.5

Required torque to lift is $< .3$ Nm

Servo stall torque is 1.73 Nm

Stall torque ratio: **5.8:1**

(operating torque remains well below motor limits)



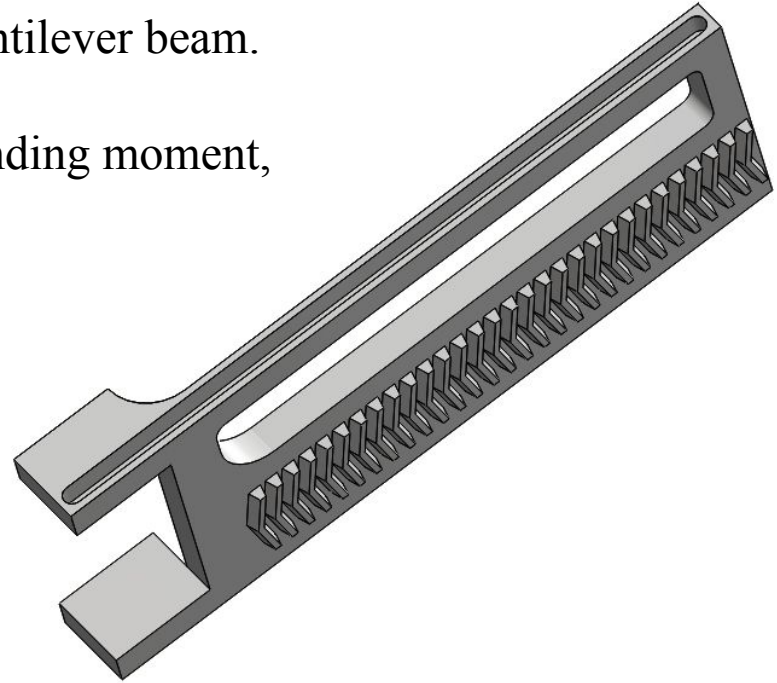
Bending forces on lift system

Due to forces on the lift it was treated like a cantilever beam.

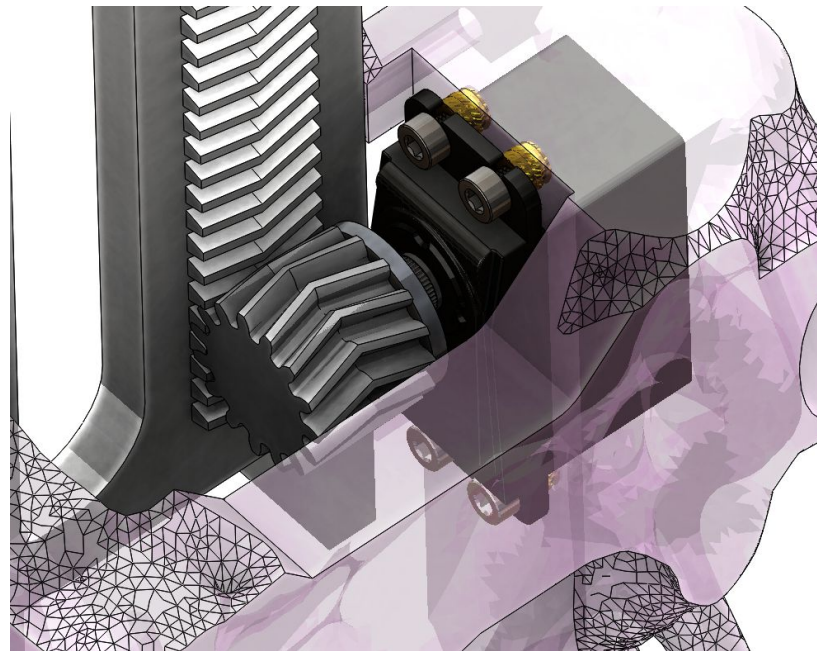
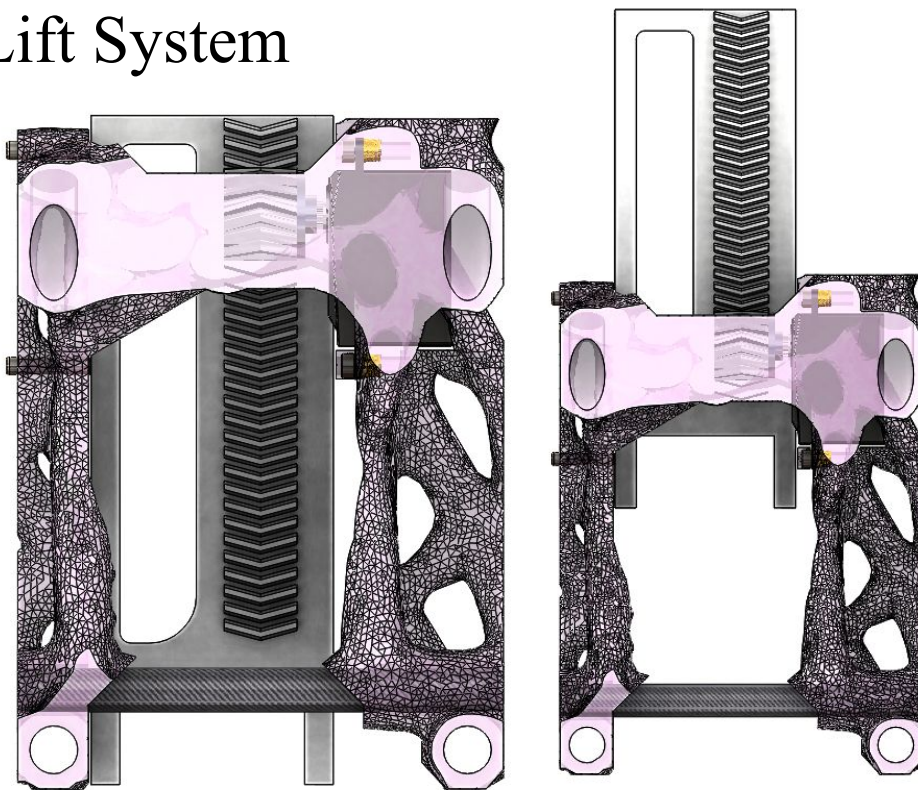
Calculated load force, torque required, shaft bending moment, resultant force, etc.

Shaft bending moment $< .202 \text{ Nm}$

F.O.S = 3 (will not bend)



Lift System





UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Chassis

Chassis

Material: PETG Filament

Mass: 118.8g

Percentage weight Front: 59%

Percentage weight Rear: 41%

Percentage weight Front Loaded: 80%

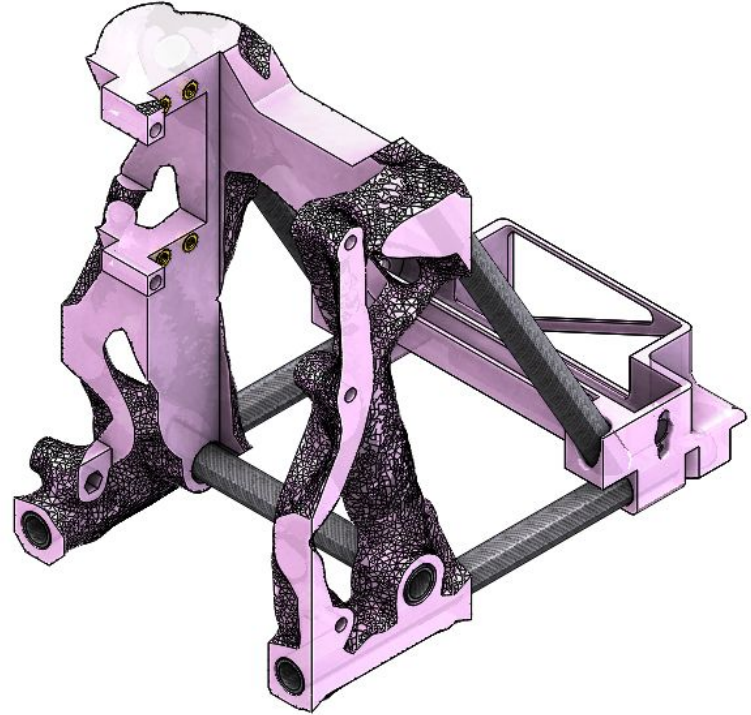
Percentage weight Rear Loaded: 19%

Deflection Loaded: 1 Micron

SF Loaded: 453

Deflection Unloaded: 0.1 Micron

SF Unloaded: 2762



Chassis

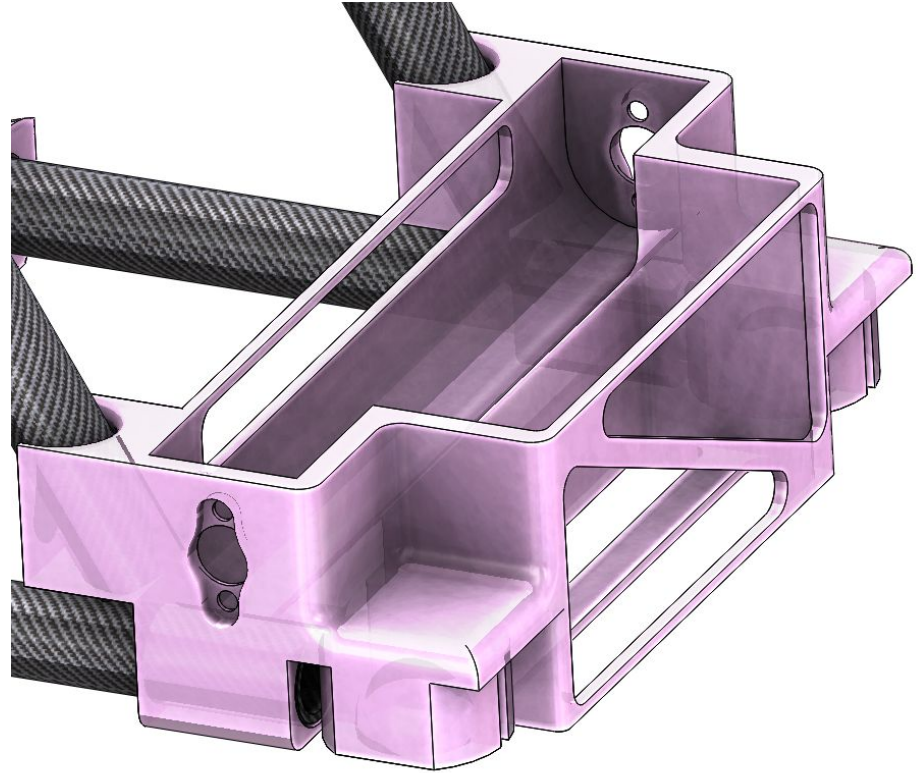
Two halves joined by carbon fiber rods
glued together

Integration with Drivetrain and Lift

Shoulder bolts for front wheel axles

Rear castors attached using a single
screw and heat set insert

Motors are attached to the chassis using
2 M2 bolts





UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Drivetrain and Steering

Drivetrain

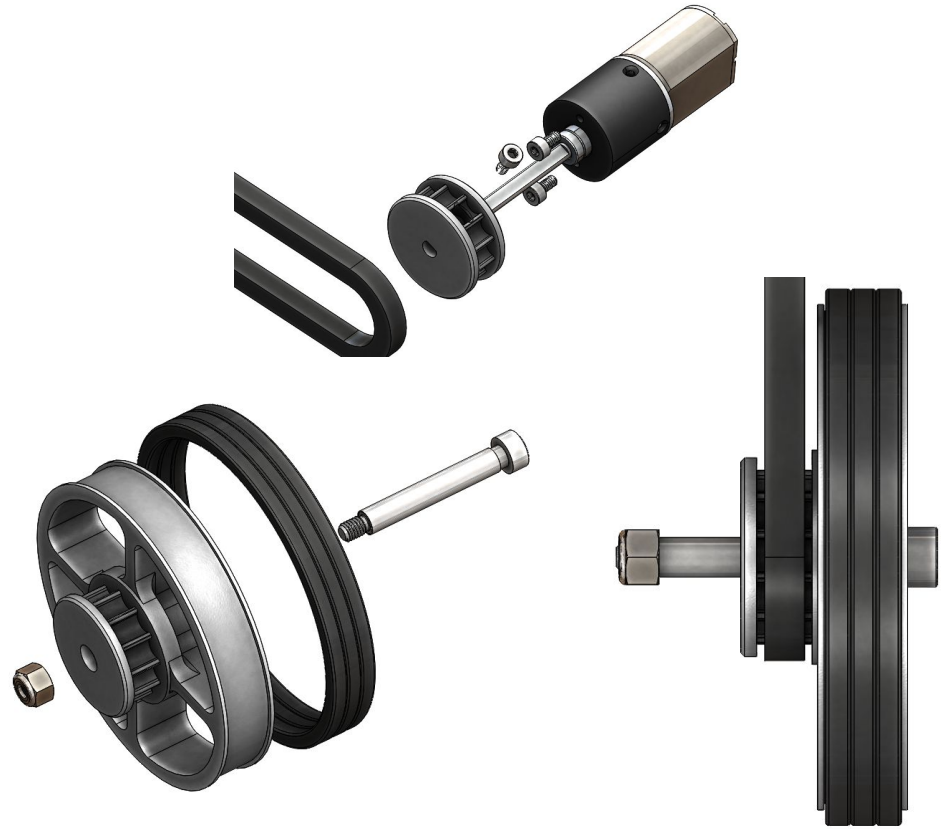
Motors: Mini Brushed Mk2

Weight: 72g

Gear Ratio: 22.6:1

Speed: 610 RPM

- Belt Drive for Rear CG
- Caster balls
 - Low Friction
 - Low Complexity



Drivetrain

Turning Moment: $0.022 \text{ N}\cdot\text{m}$

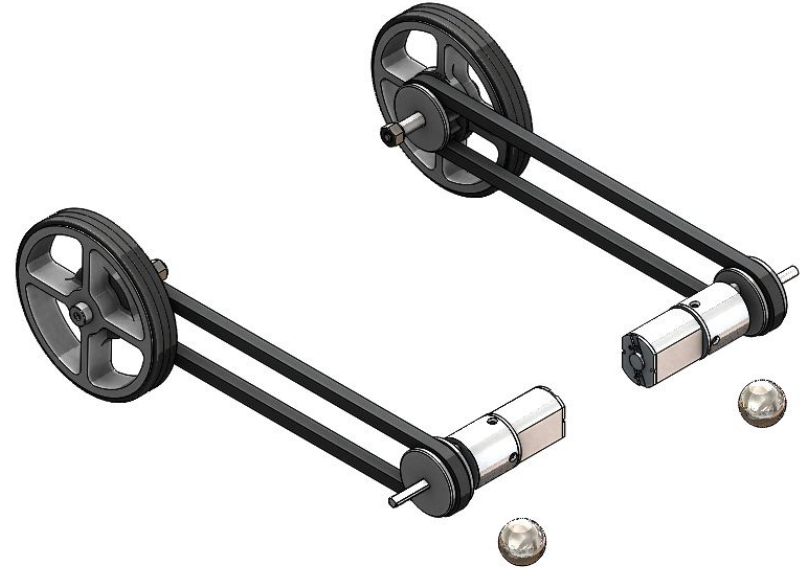
Friction Moment on Rear (Loaded):
 $4.98 \times 10^{-3} \text{ N}\cdot\text{m}$

Friction Moment on Rear (Unloaded):
 $9.51 \times 10^{-3} \text{ N}\cdot\text{m}$

Final Safety Factors:

Loaded: 4.31

Unloaded: 2.26





UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Control System

Control System Mapping

BT - Battery

C - Capacitor

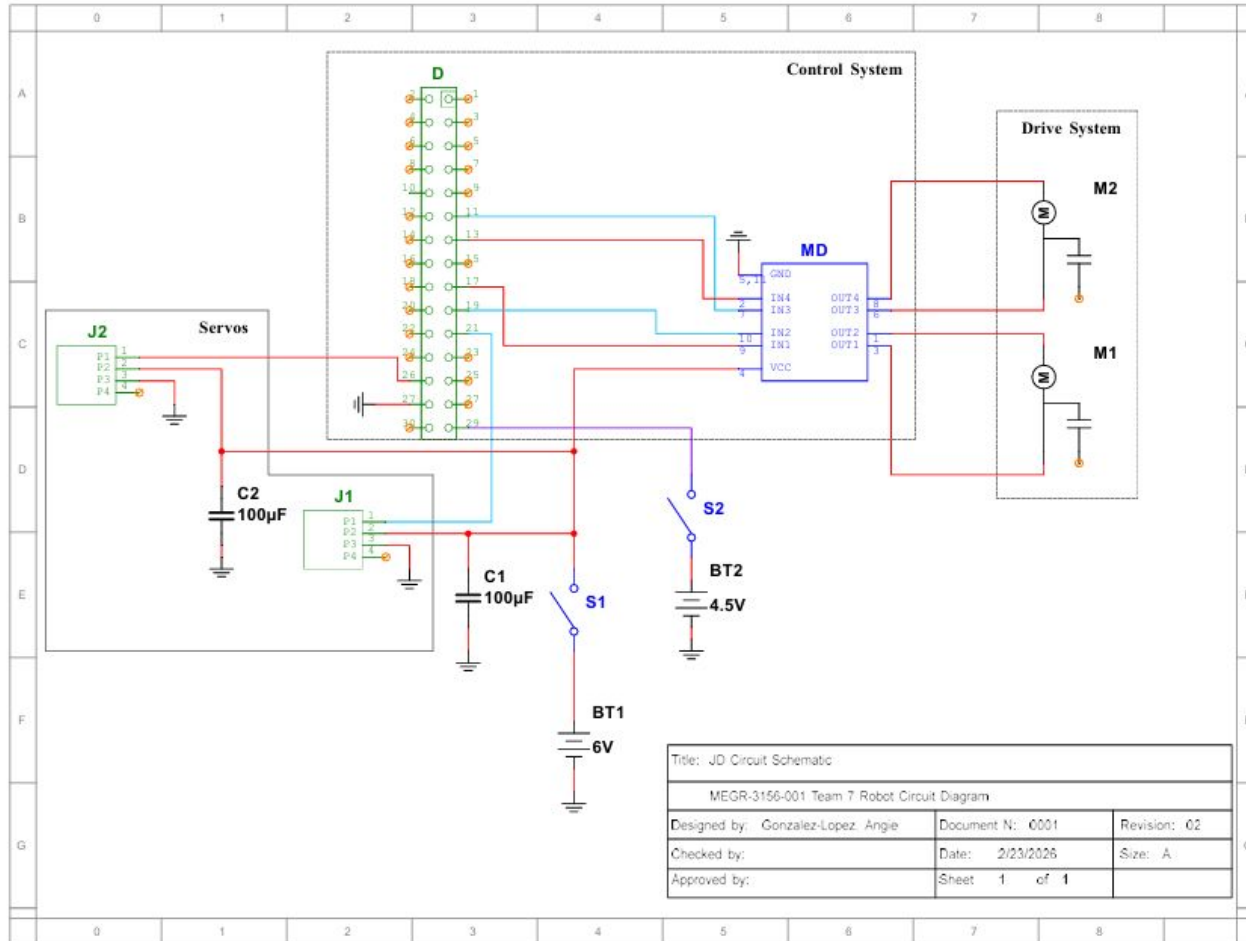
D - Development Board (ESP32)

J - INJS022 Servo

M - Mk2 Motor

MD - Motor Driver (DRV8833)

S - Switch



Control System Components

Power Supply: 280.66 g

- 7x Lithium Batteries (1.5 V each)
- 2x Battery Holder w/ Switch

Servo System: 100.094 g

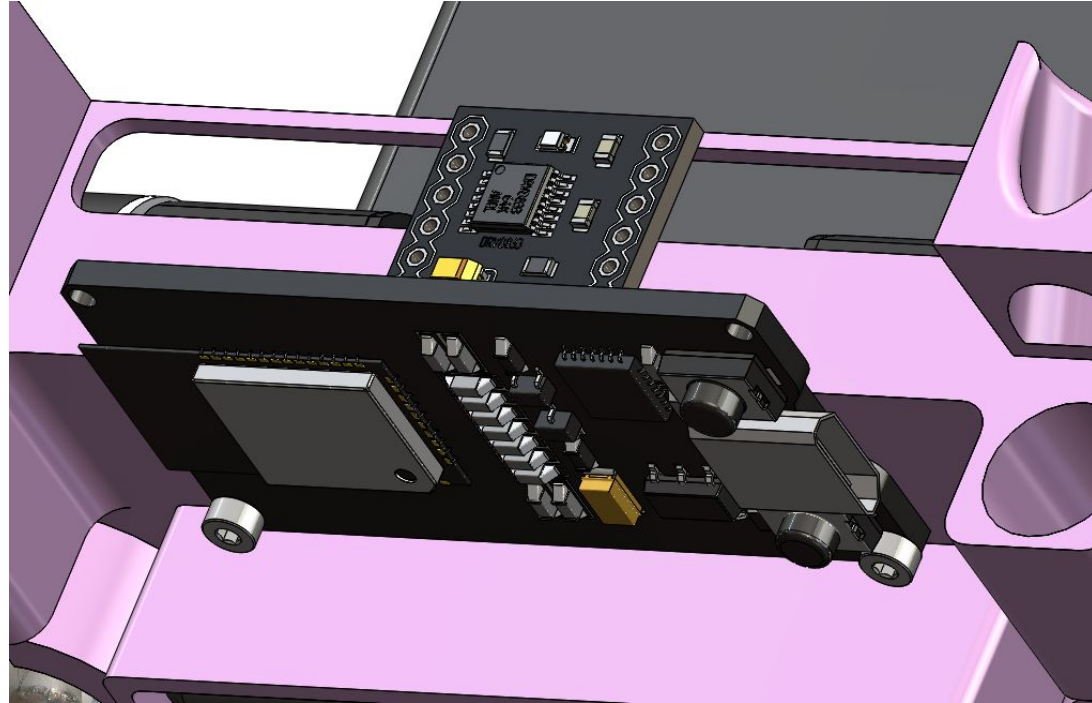
- 2x INJS022 Servos
- 2x Capacitors (100 μ F)

Drive System: 53 g

- 2x Mk2 Motors

Control System: 27.974 g

- 1x ESP32 Development Board
- 1x DRV8833 Motor Driver



Total Circuit Weight: 461.73 g

Battery Math

Battery:

$$I_{Available} := \frac{V_{Total}}{R_{Internal}} = 7.5 \text{ A}$$

$$I_{Max} := (2 \cdot I_M) + I_{CS} + I_{LS} = 2.014 \text{ A}$$

$$I_{Moving} := I_{CS} + (2 \cdot I_M) = 1.223 \text{ A}$$

$$V_{Available} := V_{Total} - (I_{Moving} \cdot R_{Internal}) = 5.022 \text{ V}$$

$$Time_{Available} := \frac{Capacity_B}{I_{Max}} = 104.285 \text{ min}$$

Theoretical Max with no losses

Significant voltage drop around 60% of battery life so having out Time available * 60% being above 30min is Ideal

$$Time_{WhileOptimal} := Time_{Available} \cdot 60\% = 62.571 \text{ min}$$

Battery Efficiency:

$$P_{loss} := I_{Moving}^2 \cdot R_{Internal} = 1.197 \text{ W}$$

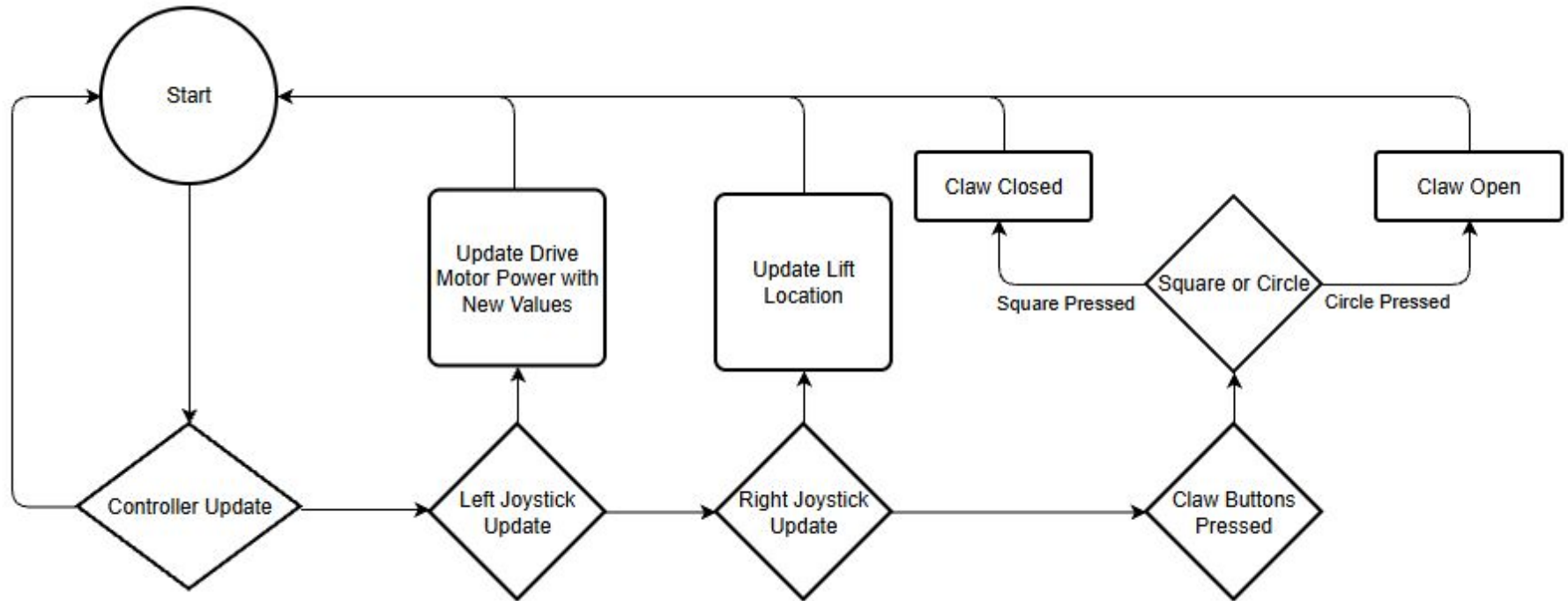
$$P_{Delivered} := V_{Available} \cdot I_{Moving} = 6.141 \text{ W}$$

$$P_{Total} := V_{Total} \cdot I_{Moving} = 7.338 \text{ W}$$

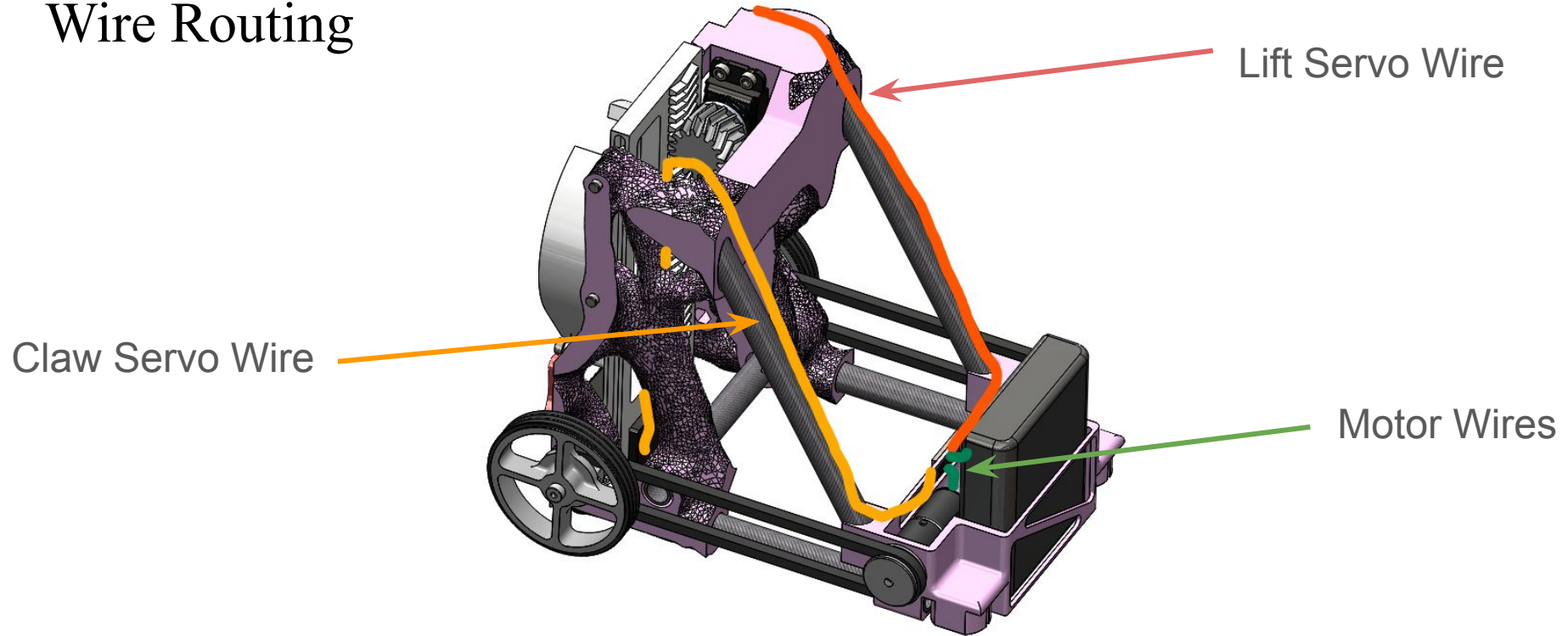
$$Efficiency := \frac{P_{Delivered}}{P_{Total}} \cdot 100 = 83.694$$

- Supply $I_{AVAILABLE} = 7.5$
Amps
- Continuous $I_{MOVING} = 1.2$
Amps
- Peak $I_{MAX} = 2.0$ Amps
- Theoretical time w/o losses
= 104 min
- Battery life = 60%, current
will experience voltage
drop.
- Efficiency = 83.7%

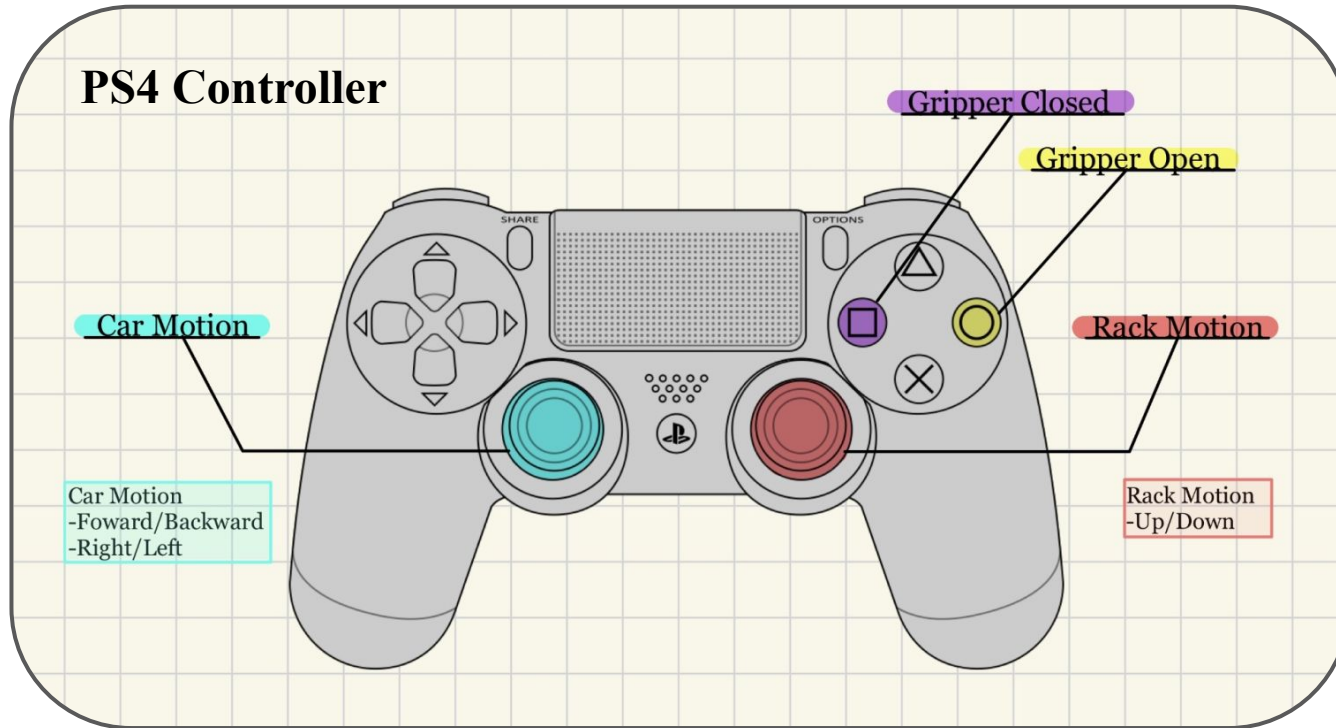
Code Flow



Wire Routing



Controller Diagram

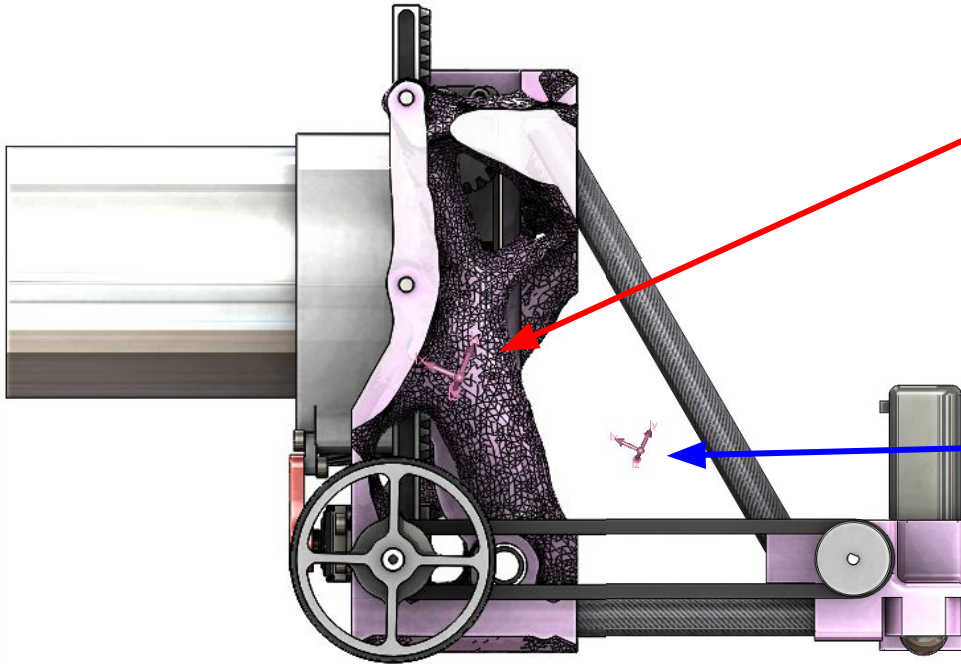




UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Overall Robot

CAD Mass Properties (Lowered)



Loaded

Mass properties of JD-2026-ChassisAssyCarbon-V1.0
Configuration: Default
Coordinate system: -- default --

Mass = 880.41902 grams
Volume = 48.97655 cubic inches
Surface area = 436.67377 square inches

Center of mass: (inches)
X = -1.68441
Y = 2.50937
Z = 0.03058

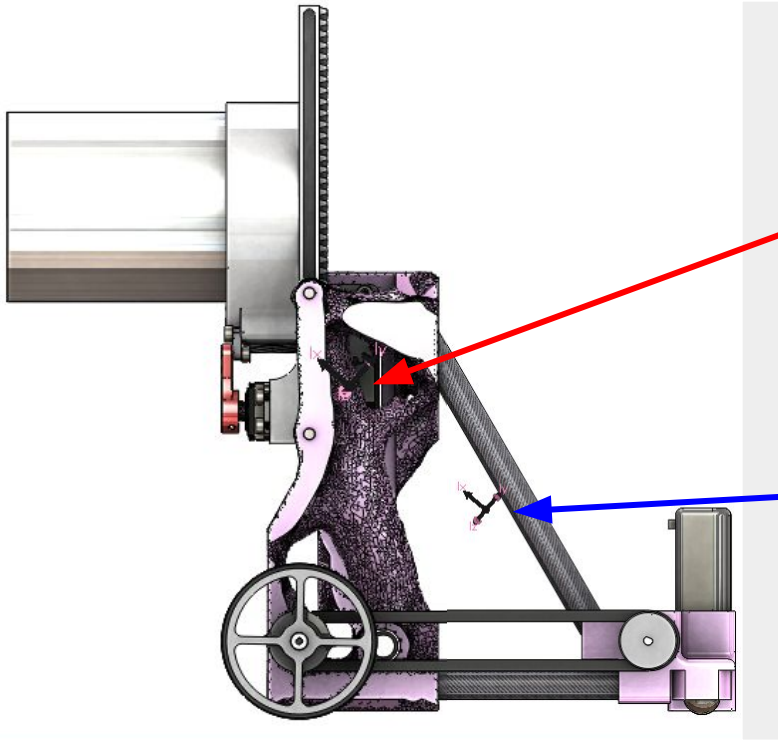
Unloaded

Mass properties of JD-2026-ChassisAssyCarbon-V1.0
Configuration: Default
Coordinate system: -- default --

Mass = 518.32017 grams
Volume = 27.78668 cubic inches
Surface area = 396.19321 square inches

Center of mass: (inches)
X = -3.56907
Y = 2.12191
Z = 0.05068

CAD Mass Properties (Raised)



Loaded

Mass properties of JD-2026-ChassisAssyCarbon-V1.0
Configuration: Default
Coordinate system: -- default --
Mass = 881.13370 grams
Volume = 49.02016 cubic inches
Surface area = 439.21761 square inches
Center of mass: (inches)
X = -1.68704
Y = 4.48962
Z = 0.02981

Unloaded

Mass properties of JD-2026-ChassisAssyCarbon-V1.0
Configuration: Default
Coordinate system: -- default --
Mass = 518.32017 grams
Volume = 27.78668 cubic inches
Surface area = 396.19321 square inches
Center of mass: (inches)
X = -3.56907
Y = 2.77395
Z = 0.05068

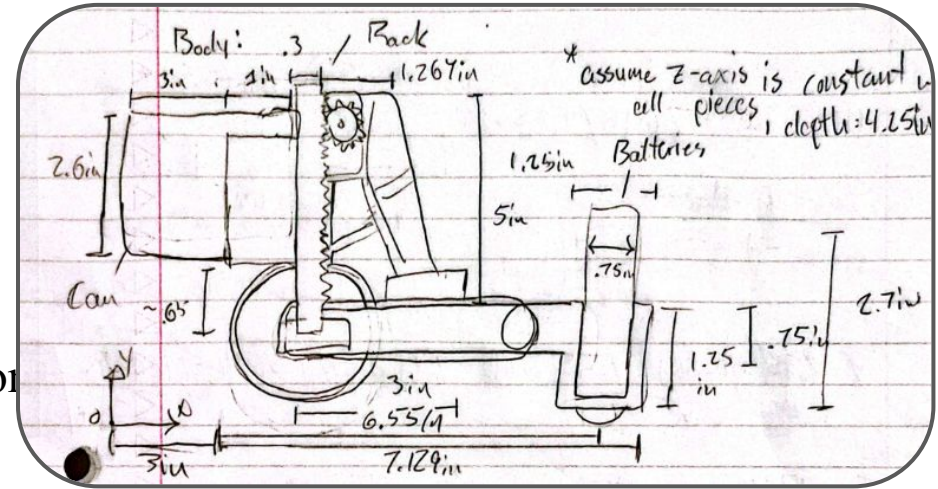


UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Stabilization and Proof

Stabilization

- Biggest obstacle is tipping whilst transporting the can.
- Max Acceleration (stall torque): 13.229 m/s^2
- Max Acceleration (slip cond.): 8.336 m/s^2
- In reality, our acceleration we decided on a desired acceleration of 0.8 m/s^2 .
- Use D'Alembert's Principle to take acceleration as a force.



Stabilization

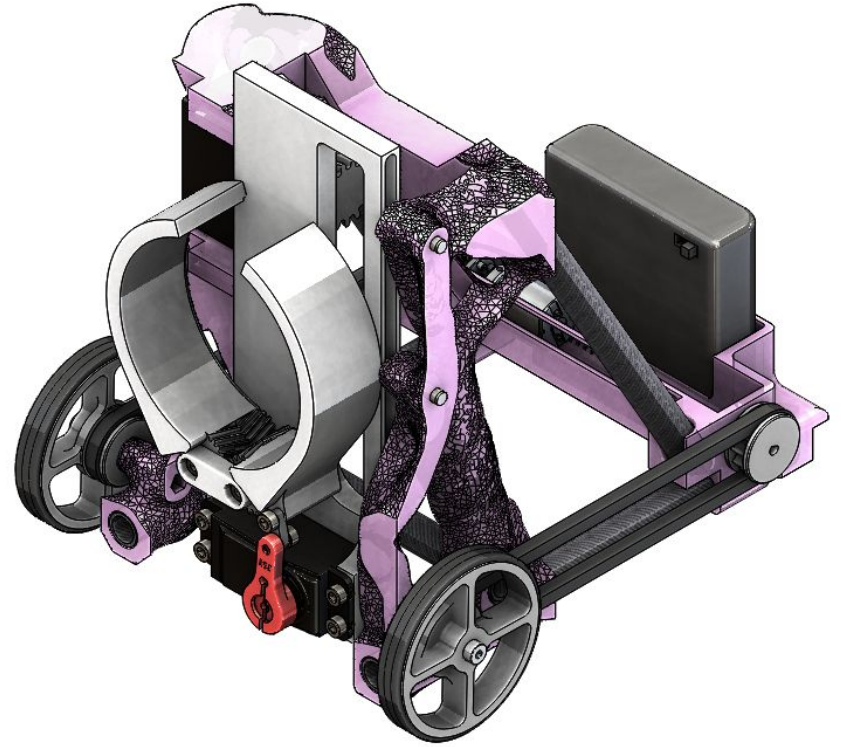
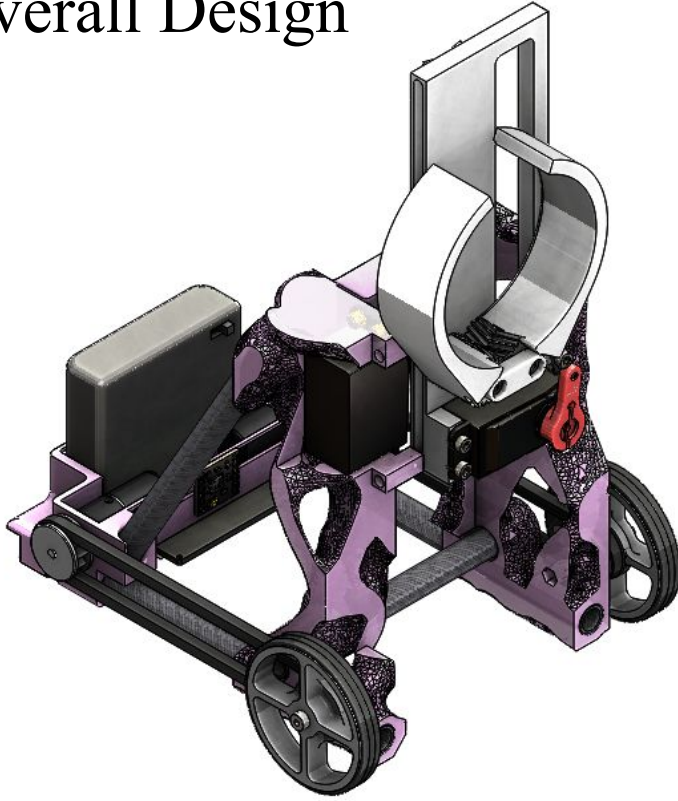
Max Acceleration (raised position):

- Back wheels pivot, inertia goes against motion.
- Summation of moment about back wheels overcomes inertia; safe.
- Safety Factor: 10.42

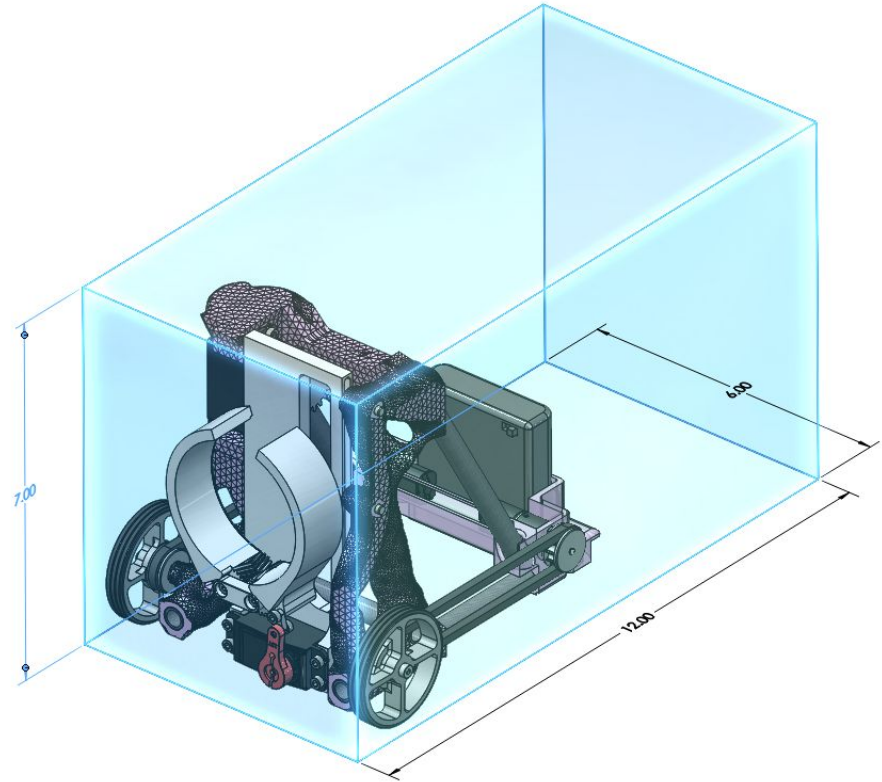
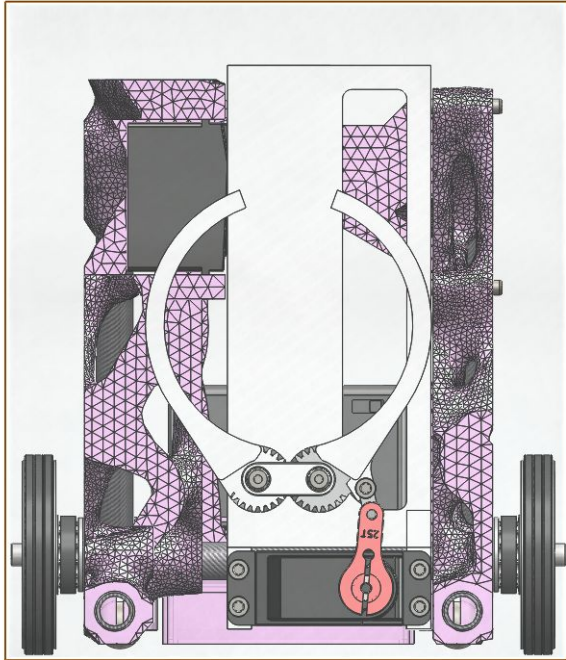
Max Deceleration (raised position):

- Front wheels pivot, inertia goes with motion.
- Summation of moment about front wheels does NOT overcome inertia; unsafe. Max: 3.677 m/s^2
- Safety Factor: 4.597

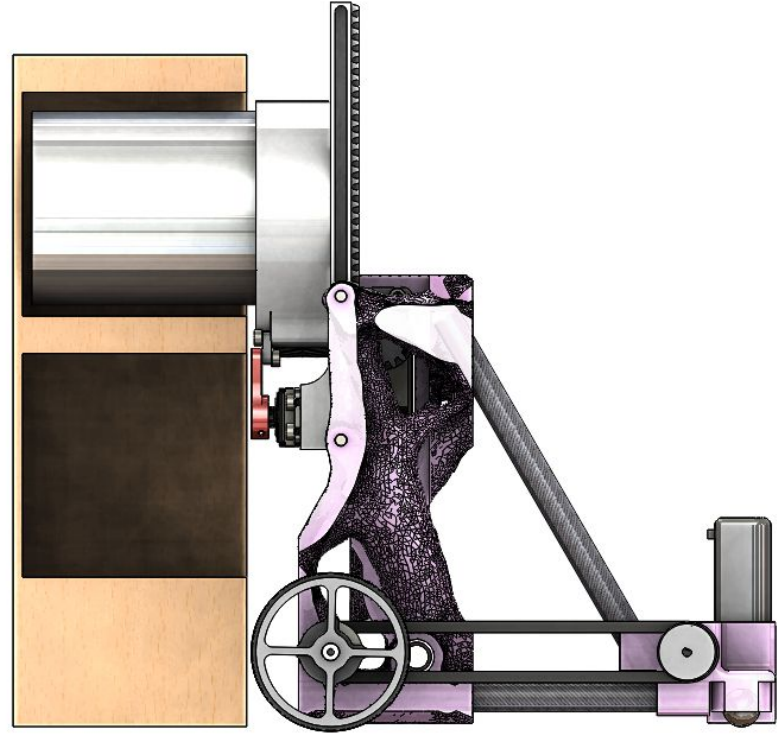
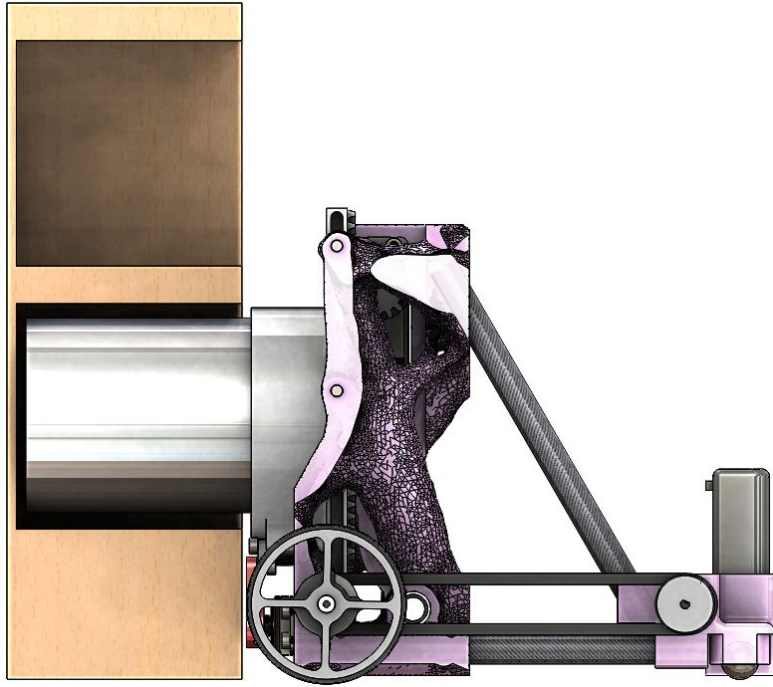
Overall Design



Dimension Check



Dimension Check



Can Transport Position

Can will be transported at the lowest cabinet slot

Height from center of can to ground: 3.5in

Center of Mass:

$X = -1.684\text{in}$

$Y = 2.509\text{in}$

$Z = 0.031\text{in}$



Anticipated Completion Time

Grip: 2 Seconds

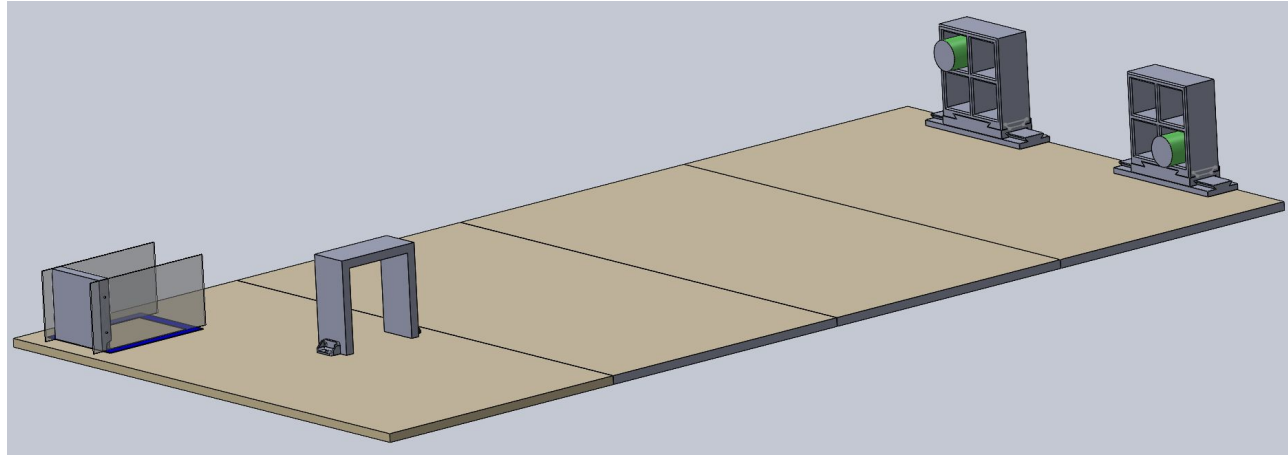
Turn: 5 Seconds

Transport: 45 Seconds

Deposit: 15 Seconds

Travel Back: 30 Seconds

Total Time: 3 Minutes





UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Use of 3D Printing

Use of 3D Printing

Part	Material	Printer	Volume	Cost
Chassis Front	PETG	Bamboo Labs A1	8.77	2.19
Chassis Rear	PETG	Bamboo Labs A1	1.82	0.46
Claw Left	PLA	Bamboo Labs A1	1.04	0.26
Claw Right	PLA	Bamboo Labs A1	1.07	0.27
Claw Linkage	PLA	Bamboo Labs A1	0.06	0.02
Lift Rack	PLA	Bamboo Labs A1	2.29	0.57
Front Wheel	PLA	Bamboo Labs A1	0.46	0.12
Rear Pulley	PLA	Bamboo Labs A1	0.11	0.03
Lift Pinion	PLA	Bamboo Labs A1	0.27	0.07



UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

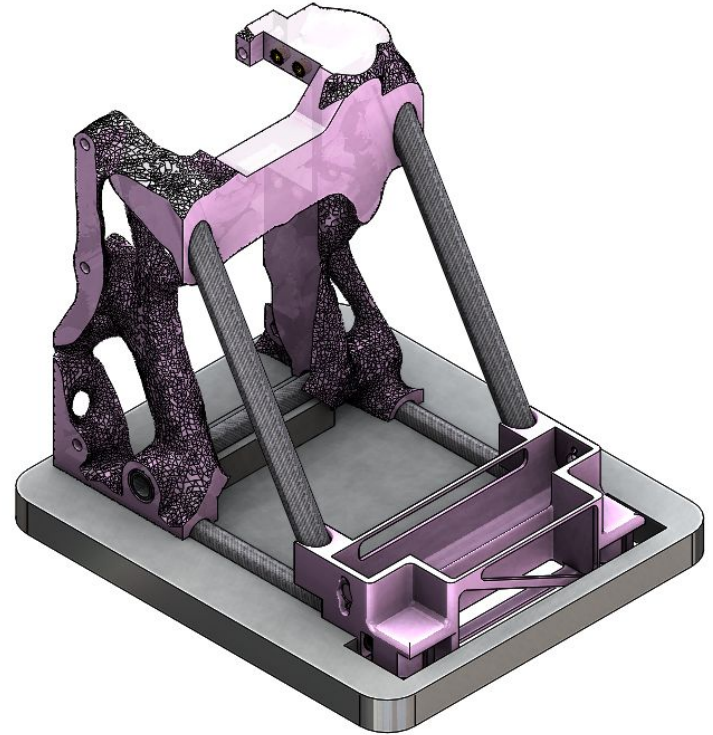
Manufacturability of the Robot

Manufacturability

- All components are easily sourceable from websites with predictable lead times
- The assembly of the main components of the robot follow a simple process that is repeatable
- The assembly and manufacturing of the the different components was take into account during the design phase
- Hand Manufactured parts where kept simple and without high tolerance

Chassis Layup

- Due to the way that the chassis is assembled that is the possibility for the chassis to be assembled crooked
- To solve this problem a alignment jig will be used to ensure that during the initial layup period the chassis will be assembled square





UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Budgets/Bill of Materials/Bill of Labor

Bill of Materials

Total: \$149.68

Chassis: \$15.88

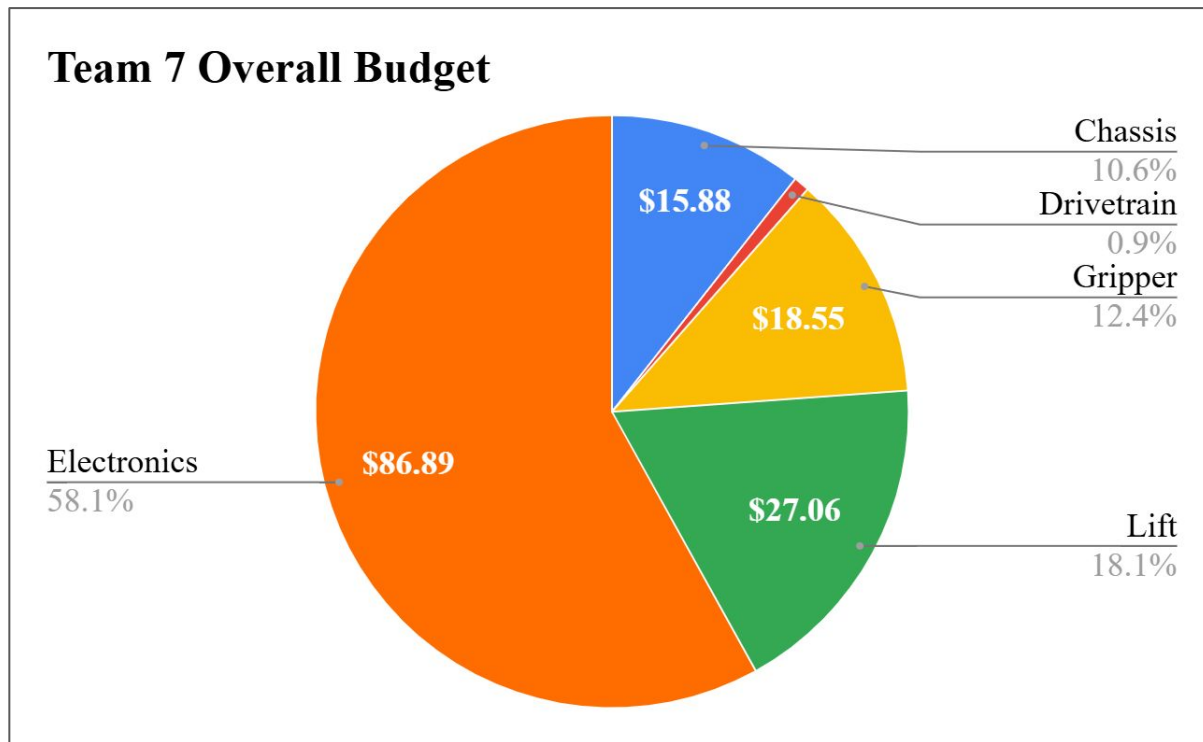
Electronics: \$86.89

Drivetrain: \$1.29

Gripper: \$18.55

Lift: \$27.06

Hardware: \$18.00 (exempt)



Bill of Labor

Team 7 - Bill of Labor			
Category	Task	Hours	Cost
Estimated Fabrication Hours			
	Machining Carbon Tubes	1	\$ 10.00
	Machining Lift Pins	1	\$ 10.00
	3D Printer Setup	1	\$ 10.00
Estimated Assembly Hours			
	Robot Assembly	3	\$ 30.00
Estimated Testing Hours			
	Robot Testing	4	\$ 40.00
Total Cost			\$ 100.00

Team 7 Bill of Labor



- Estimated Fabrication Hours
- Estimated Assembly Hours
- Estimated Testing Hours



UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

Summary of Revisions to Address First Design Review

Summary of Revisions - Gripper

Gripper Gear Pitch

- $0.2159 \times 10^{-2}\text{m}$

Gear Deflection

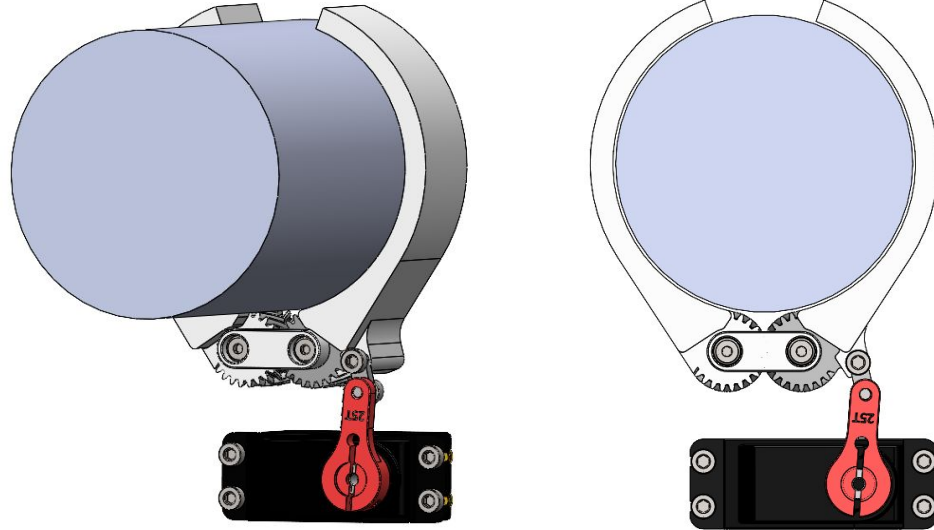
- Support Cross Member

Gripper Deflection

- $2.97 \times 10^{-6}\text{m}$

Gripper Safety Factor

- FOS: 3.95

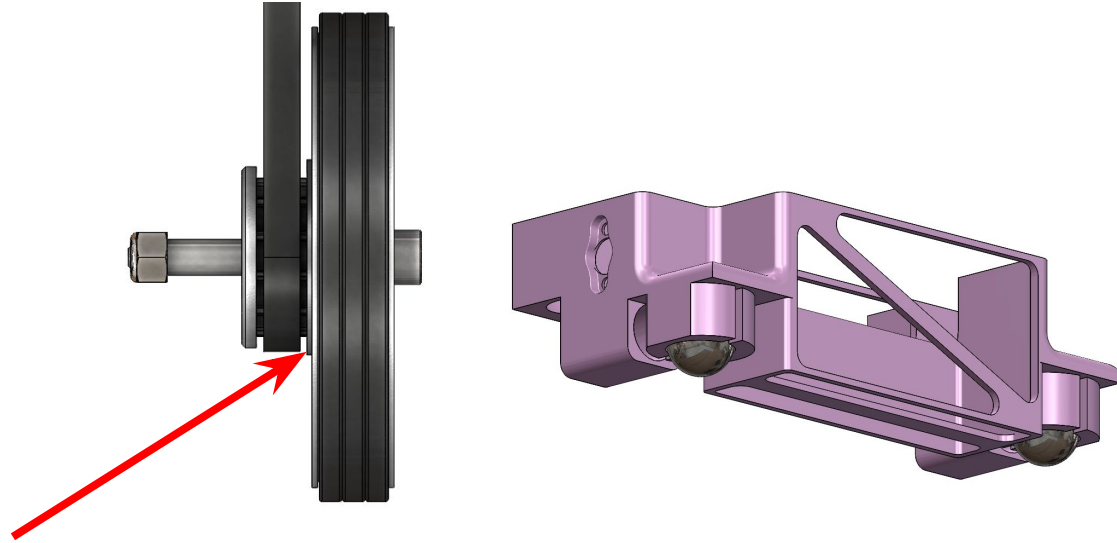


Summary of Revisions - Caster Balls / Drive Pulley Configuration

- Added clearance area between belt and wheel to reduce possible friction (Red Arrow)
- Rear caster ball material changed with consideration to weight, and safety factors calculated

Unloaded: 2.56

Loaded: 4.31



Summary of Revisions - Lift System

Force on dowel pins was not sufficient enough to bend the lift. No struts needed.

Herringbone configuration of teeth will help to improve gear alignment and keep pinion engaged.

Safety factors are >3

System was validated mechanically by comparing lifting torque to available servo torque.

Structurally validated by comparing bending stress to material yield strength.

Summary of Revisions - Battery

- Fixed internal resistance by using 0.8 Ohms, 0.2 Ohms per battery
- Added in the expected voltage drop at around 60% of the battery life
- Factored the loss of power due to the internal heat generation
- Fixed Current values by using correct values for the calculation

Battery:

$$I_{Available} := \frac{V_{Total}}{R_{Internal}} = 7.5 \text{ A}$$

$$I_{Max} := (2 \cdot I_M) + I_{CS} + I_{LS} = 2.014 \text{ A}$$

$$I_{Moving} := I_{CS} + (2 \cdot I_M) = 1.223 \text{ A}$$

$$V_{Available} := V_{Total} - (I_{Moving} \cdot R_{Internal}) = 5.022 \text{ V}$$

$$Time_{Available} := \frac{Capacity_B}{I_{Max}} = 104.285 \text{ min}$$

Theoretical Max with no losses

Significant voltage drop around 60% of battery life so having out Time available * 60% being above 30min is Ideal

$$Time_{WhileOptimal} := Time_{Available} \cdot 60\% = 62.571 \text{ min}$$

Battery Efficiency:

$$P_{loss} := I_{Moving}^2 \cdot R_{Internal} = 1.197 \text{ W}$$

$$P_{Delivered} := V_{Available} \cdot I_{Moving} = 6.141 \text{ W}$$

$$P_{Total} := V_{Total} \cdot I_{Moving} = 7.338 \text{ W}$$

$$Efficiency := \frac{P_{Delivered}}{P_{Total}} \cdot 100 = 83.694$$

Summary of Revisions - Chassis

Issues:

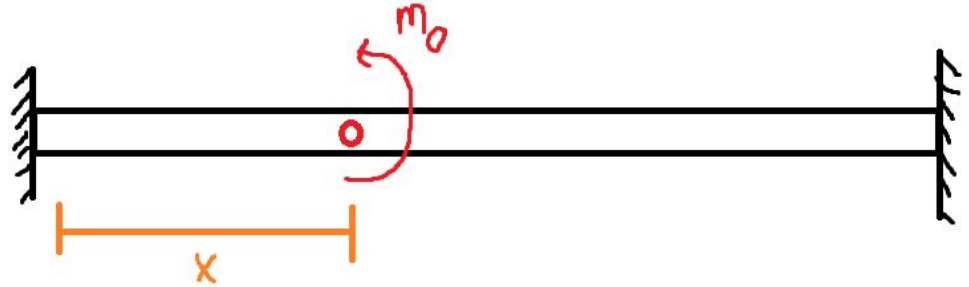
Needed Deflection and SF Values

Solution:

Used a simply supported beam to be able to calculate the deflection from the applied moment as a location x away from the front axle.

Then found the safety factor using a max deflection of 0.5mm

$$Deflection := \frac{M_o}{6 E \cdot I \cdot L} \left(\frac{L^2 - 3 b^2}{3} \right)^{\frac{3}{2}} = (1.102 \cdot 10^{-3}) \text{ mm}$$
$$SF := \frac{0.5 \text{ mm}}{Deflection} = 453.529$$





UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

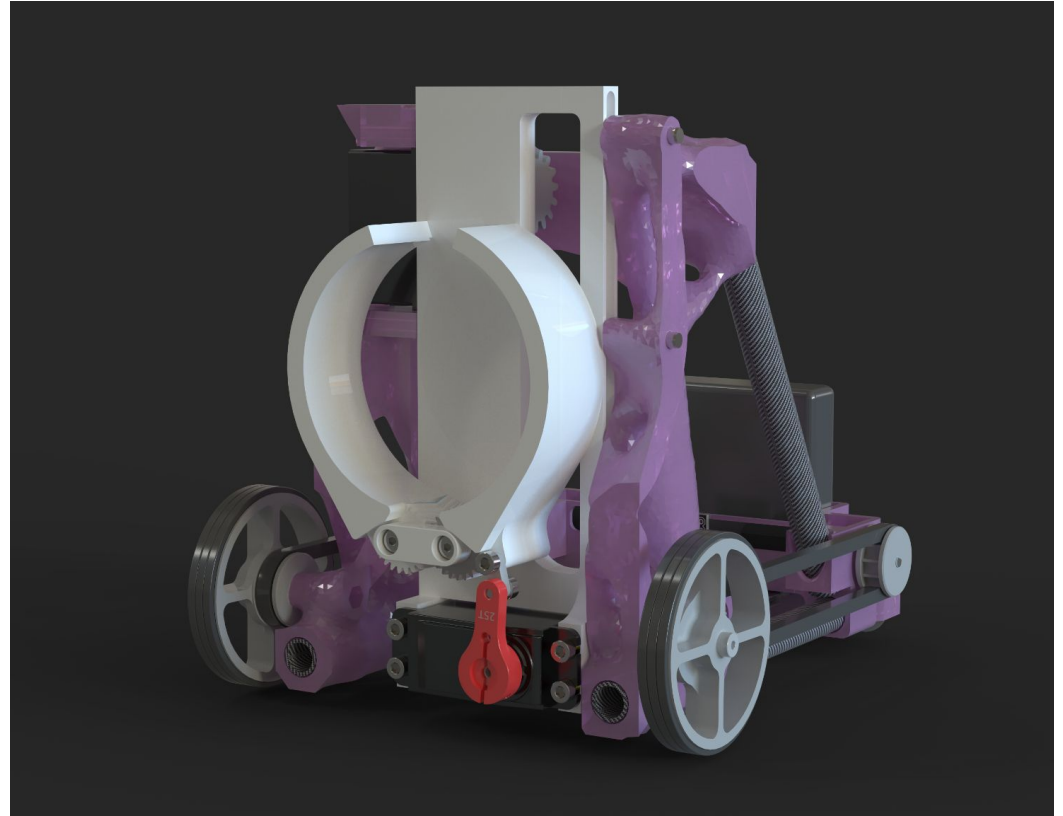
Final Summary

Final Summary

Weight: 1.14lbs

Cost: \$149.68

- Rack and Pinion Lift
- Belt Drive
- Rear Mounted Motors
- Carbon Fibre Tube Chassis



Sources used in Presentation

“PowerPoints - Brand Identity & Visual Standards.” 2023. Brand Identity & Visual Standards. April 19, 2023. <https://brand.charlotte.edu/visual-identity/powerpoints>.